# Welcome to
# **Intro to NetCDFs with *QGIS***

Questions to think about:

- What is your name?
- What program/department are you from?
- What brought you to the workshop today?
- What do you hope to get out the workshop?

2026

University of Victoria | Libraries

# Welcome to
# **Intro to NetCDFs with *QGIS***

it is **highly recommended** that anyone attempting this workshop **first go through**:

- Intro to Vector Data with *QGIS*

https://uviclibraries.github.io/qgis-vector

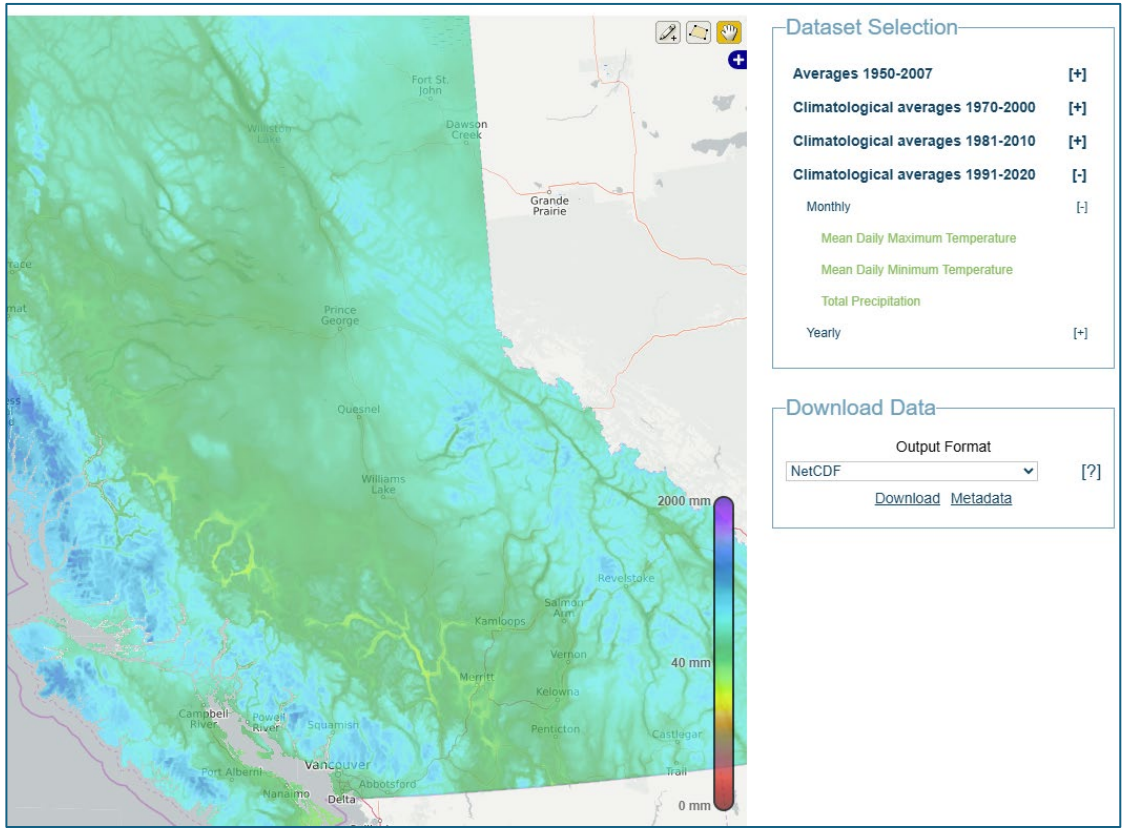
- Intro to Raster Data with *QGIS*

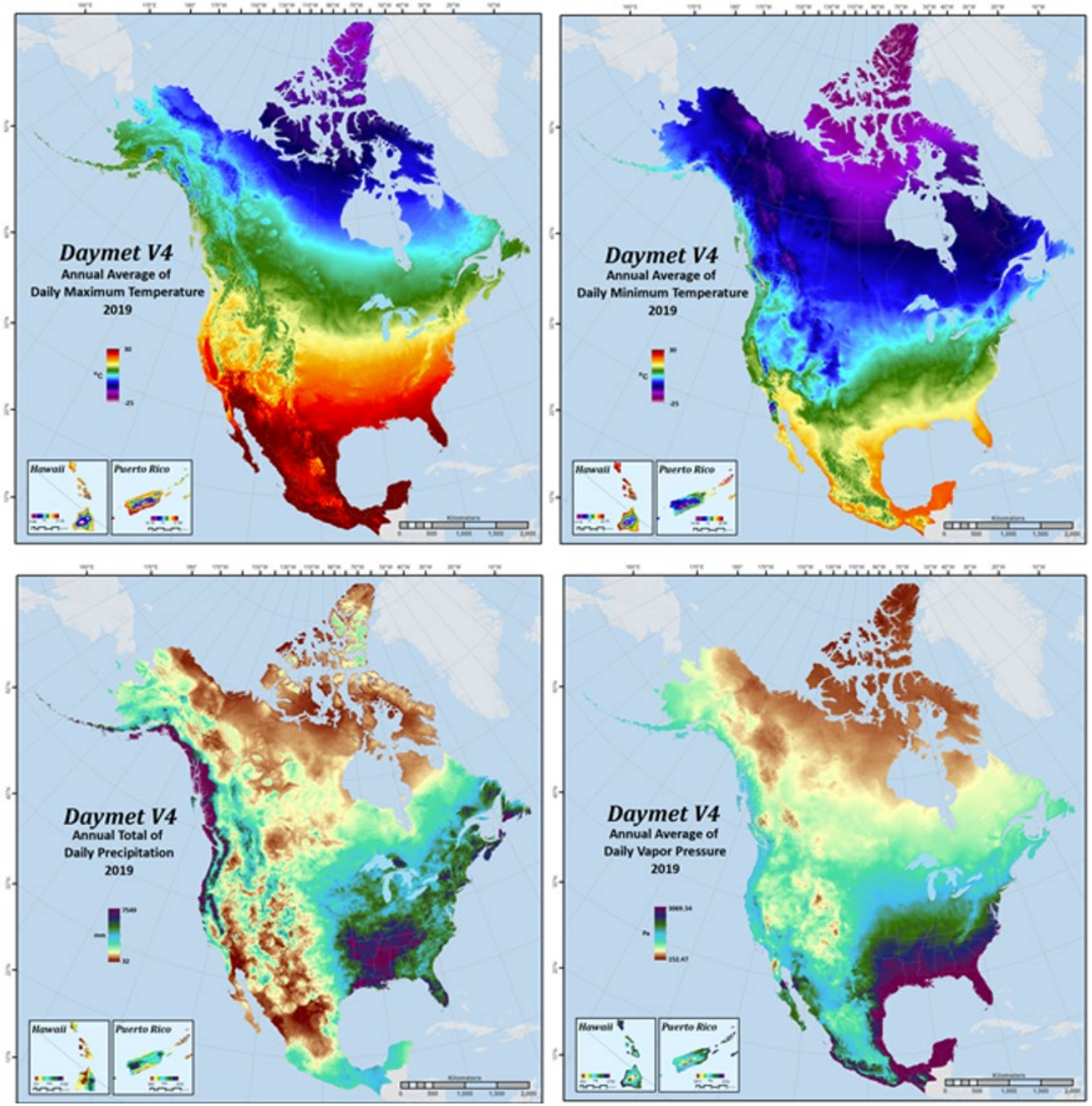
https://uviclibraries.github.io/qgis-raster

2026

University of Victoria Libraries

# Intro to NetCDFs with *QGIS*



## Learning Objectives

- define NetCDFs
- describe why/where NetCDFs are used
- explain why/where NetCDFs are sometimes difficult to use

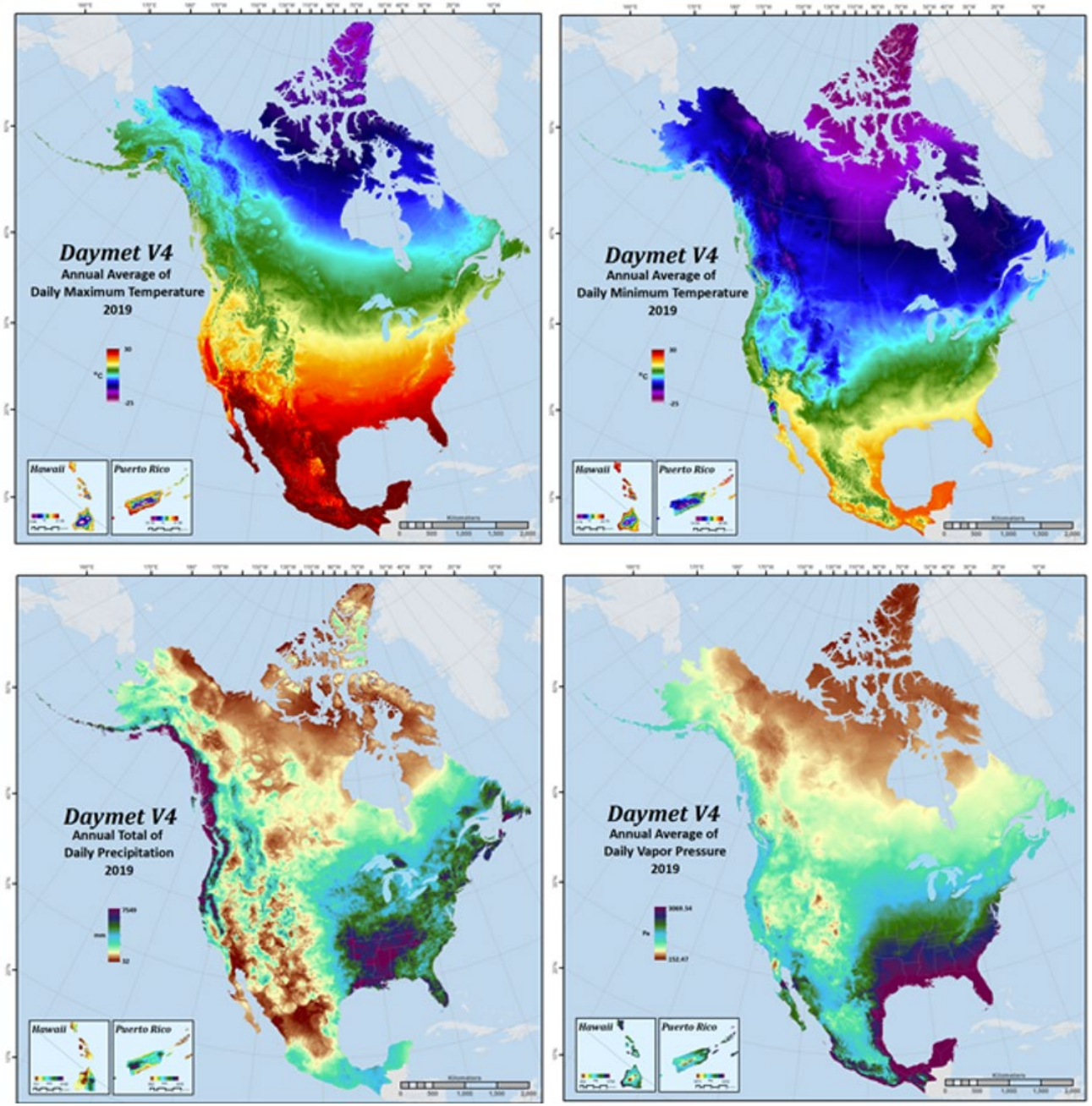
**Outcomes** (of this workshop)

- load two different NetCDFs
- Symbolize NetCDF layers
- Examine NetCDF layers
- utilize vector polygons to determine precipitation rates in BC
  - (with 2023 *BC Fire Perimeters* and *QGIS* Zonal Statistics)

**NOTE**: this is NOT a statistical analysis of forest fires in BC…
…just utilizing NetCDFs…

why NetCDFs?

common file format…
for climate/weather data…

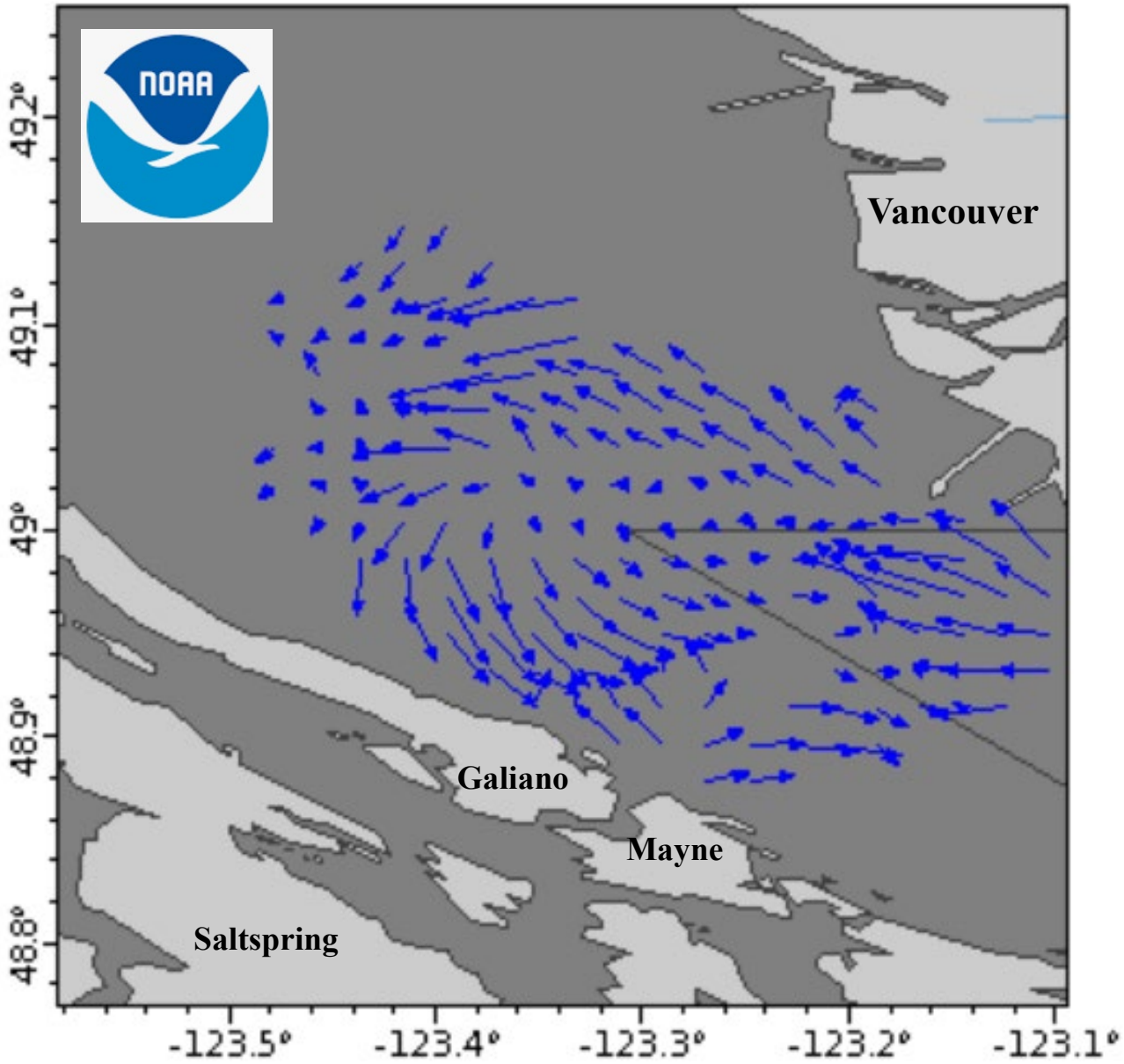



DAYMET
DAILY SURFACE WEATHER AND CLIMATOLOGICAL SUMMARIES
ORNL DAAC

*Daymet* Documentation / Methods
https://daymet.ornl.gov/
https://www.earthdata.nasa.gov/data/projects/daymet

why NetCDFs?

common file format…
for oceanic data…




*Currents:* HF Radar, US West Coast, RTV, NRT, Hourly, 2012-present, 1km

https://coastwatch.pfeg.noaa.gov/erddap/griddap/ucsdHfrW1.html

# why NetCDFs:

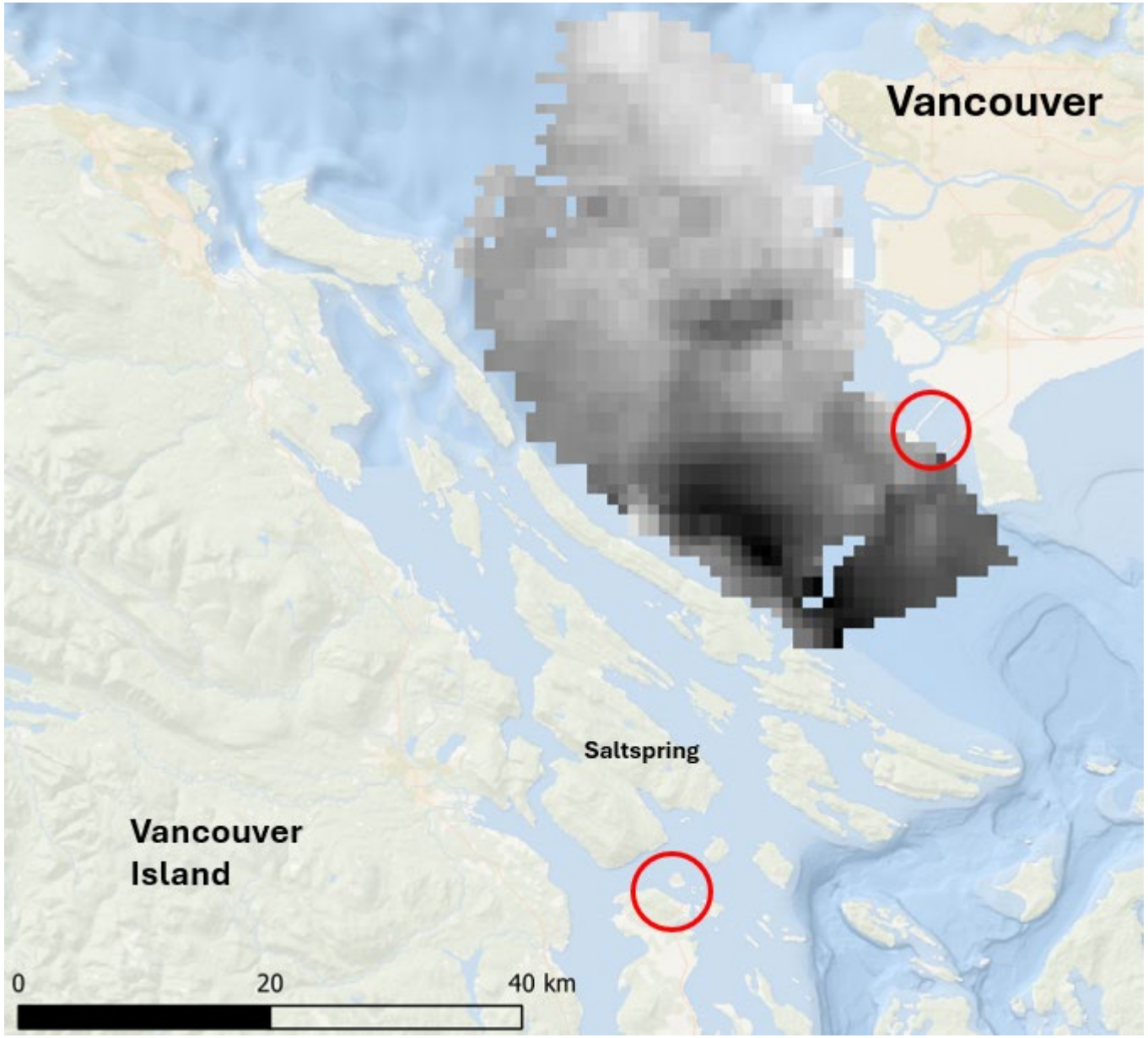
**multiple:**

- variables
- time dimensions

**cells with**

- Lat
- Long
- u-Current
- v-Current
- u-wind
- v-wind
- etc

- **Date/Time**




*Currents:* HF Radar, US West Coast, RTV, NRT, Hourly, 2012-present, 1km

https://coastwatch.pfeg.noaa.gov/erddap/griddap/ucsdHfrW1.html BaseMap: *ESRI Ocean* via *QGIS*

SalishSeaCast ERDDAP™
Public access to the SalishSeaCast model products

Forecast, Salish Sea, 2d Wave Fields, 30min, v17-02

example of multiple variables

ucur (Eastward Current, m s-1)
vcur (Northward Current, m s-1)
uwnd (Eastward Wind, m s-1)
vwnd (Northward Wind, m s-1)
hs (Significant Height of Wind Waves, m)
lm (Mean Wave Length, m)
t02 (Mean Period T02, s)
fp (Wave Peak Frequency, s-1)
dir (Wave Mean Direction, degree)
dp (Peak Direction, degree)
wcc (Whitecap Coverage, 1)
wch (Significant Breaking Wave Height, m)
utwo (Eastward Wave to Ocean Stress, m2 s-2)
vtwo (Northward Wave to Ocean Stress, m2 s-2)
foc (Wave to Ocean Energy Flux, W m-2)
uuss (Eastward Surface Stokes Drift, m s-1)
vuss (Northward Surface Stokes Drift, m s-1)

**File type:** (more information)

.nc - Download a NetCDF-3 binary file with COARDS/CF/ACDD metadata.

https://salishsea.eos.ubc.ca/

**SalishSeaCast ERDDAP™**
Public access to the SalishSeaCast model products

Forecast, Salish Sea, 2d Wave Fields, 30min, v17-02

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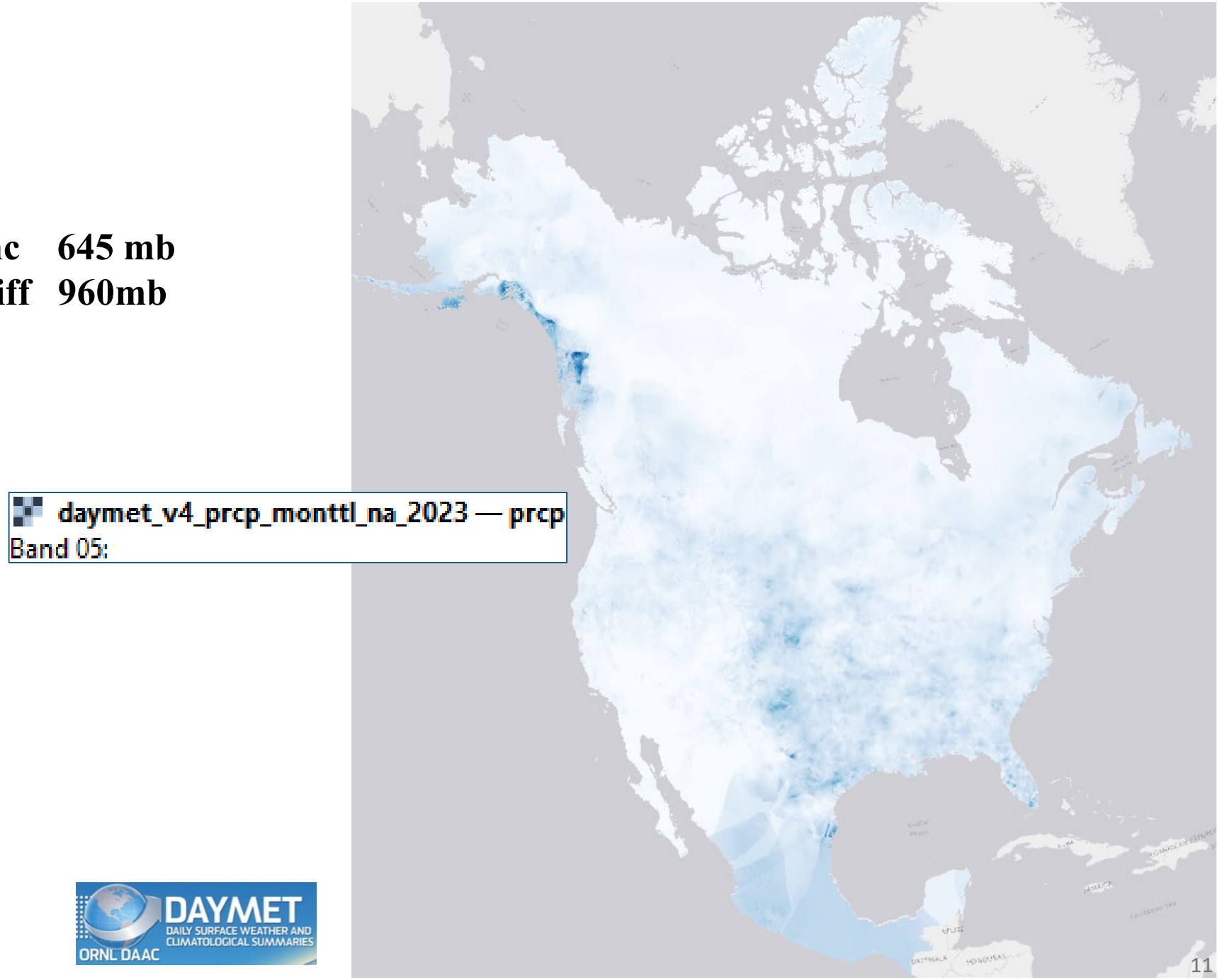
https://salishsea.eos.ubc.ca/
via *QGIS*

# why NetCDFs?

*daymet_v4_prcp_monttl_na_2023*.**nc** **645 mb**
*daymet_v4_prcp_monttl_na_2023*.**tiff** **960mb**

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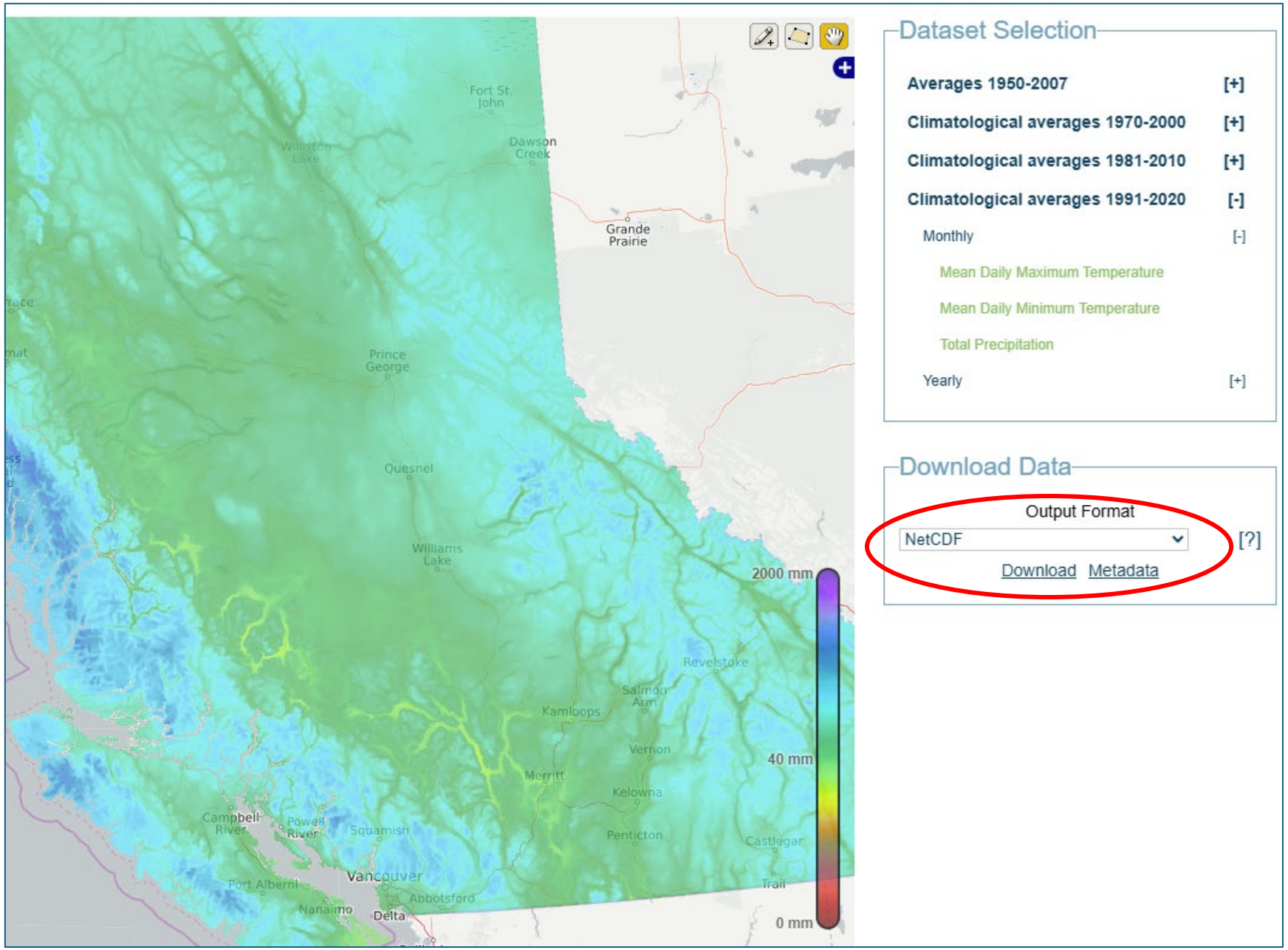
common file format…




PACIFIC CLIMATE IMPACTS CONSORTIUM

*PRISM Monthly Climatology* Documentation / Methods
https://www.uvic.ca/pcic/data-analysis-tools/data-portal/finescale-gridded-climatologies/index.php
https://data.pacificclimate.org/portal/bc_prism/map/

**BUT…**

**“There's nothing I hate more than NetCDF files.”**

**“They consistently fail at being at the right place / in the right projection.”**
https://www.reddit.com/r/gis/comments/17uxh27/theres_nothing_i_hate_more_than_netcdf_files_they/ 2024

**“unable to fix the projection of NetCDF file”** 2025

**“NetCDF files are projected incorrectly in *ArcGIS Pro*”** 2025

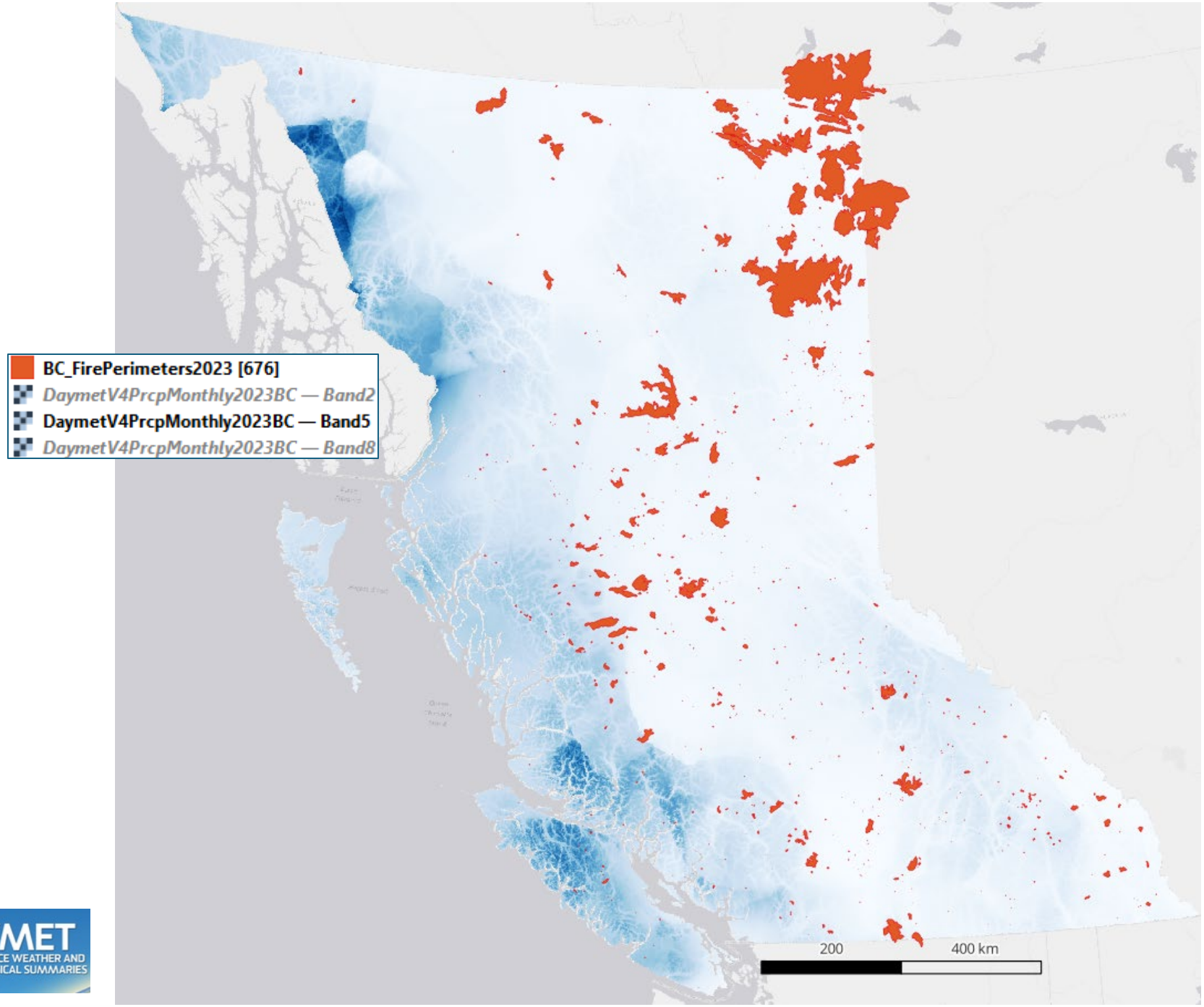
**caveat:** NetCDFs can be difficult to work with GIS…

…this workshop might give a false impression of ease…

this workshop will utilize **grid NetCDFs**…(with Lat/Long coordinates )
…in contrast to table NetCDFs

# 2023 Monthly Precipitation

*daymet_v4_prcp_monttl_na_2023*.**nc**

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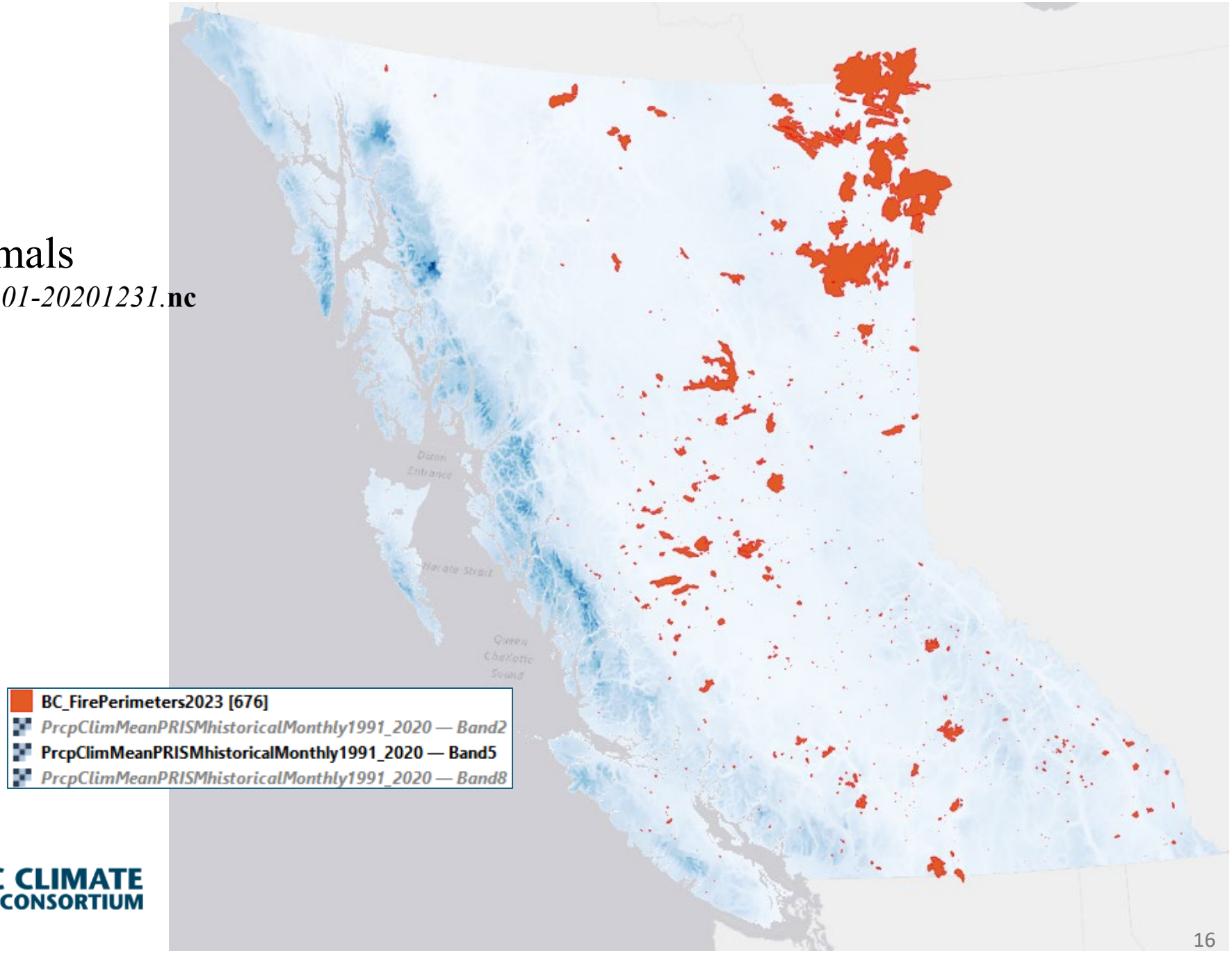
# 1990-2020 Precipitation Normals

*pr_mClimMean_PRISM_historical_19910101-20201231.***nc**




PACIFIC CLIMATE IMPACTS CONSORTIUM

# Understand *QGIS* interface

*your interface may look different

Menu Bar

Toolbar

Layers Panel

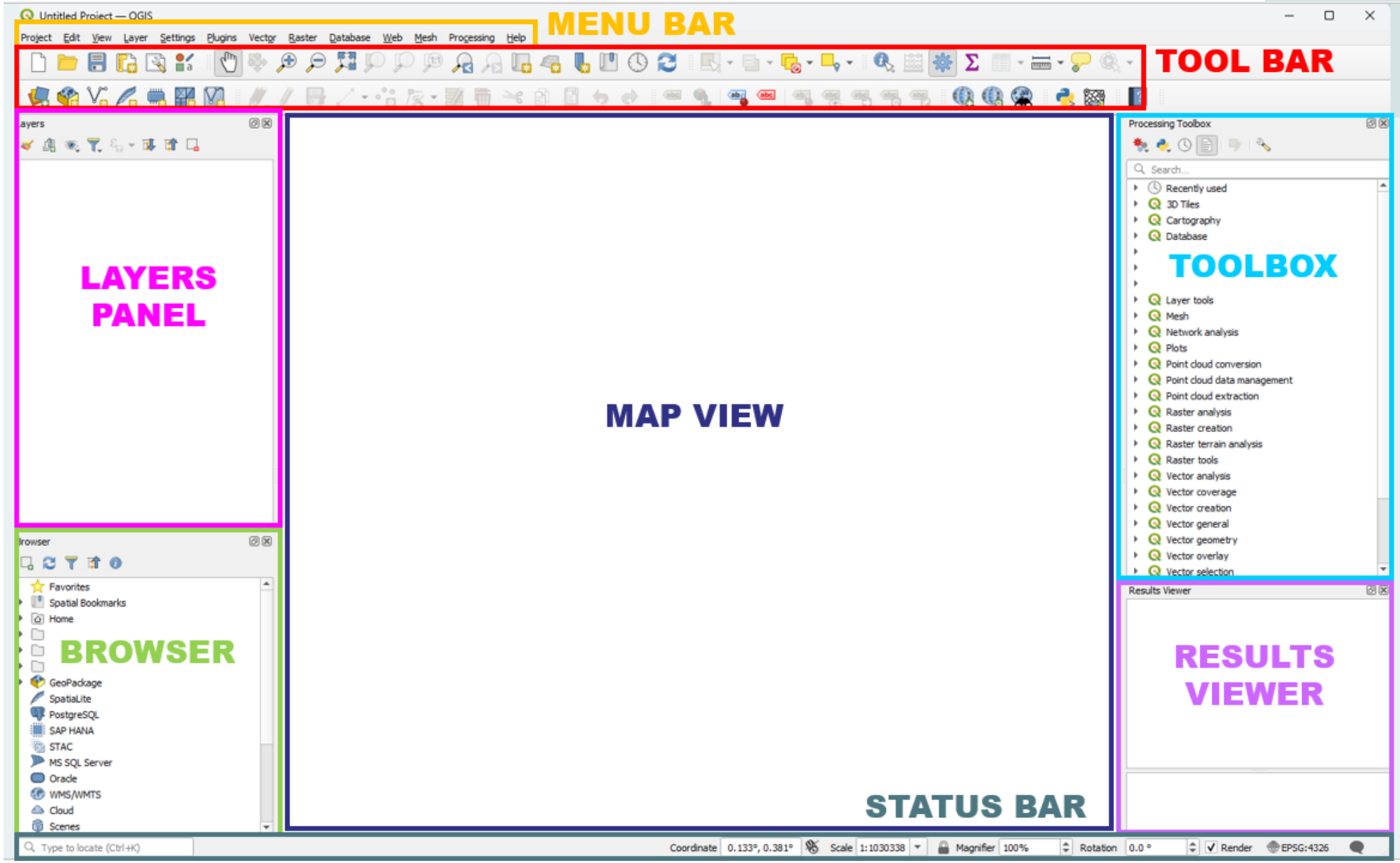
Map view

Browser

Toolbox

Status Bar

Results Viewer



# Understand *QGIS* interface

**Menu Bar:**


Horizontal bar at the top providing access to various functions and tools

*(Project management, Edit, Plugins, Vector & Raster tools, etc.)*

**Toolbar:**

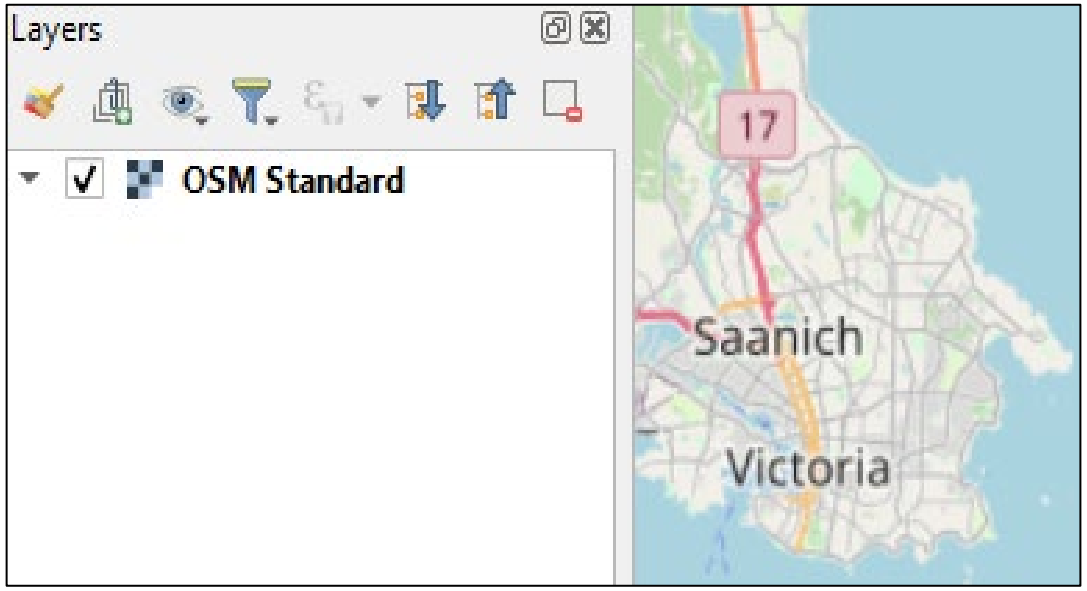
Contains icons for frequently used tools, such as Add Data, Pan, Zoom, Identify, etc.

Quick access to essential operations.

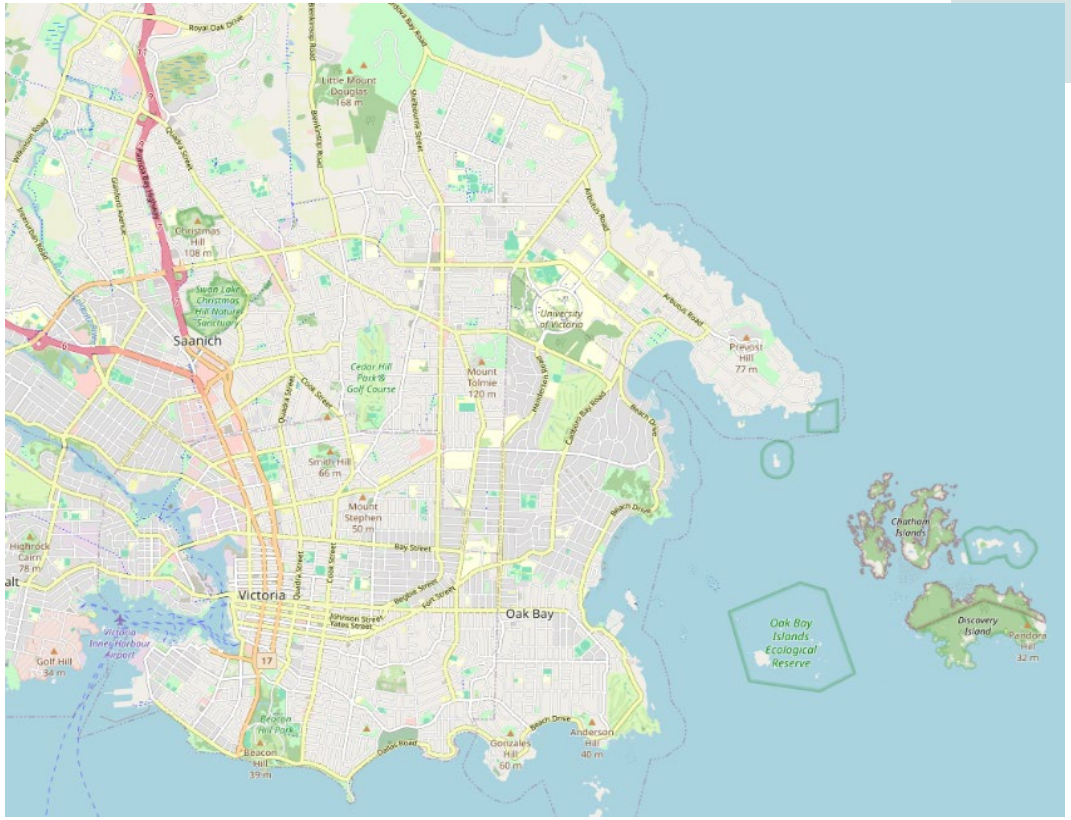
**Layers Panel:**

**Displays** all active **Layers** in the project.

Allows users to organize, manage visibility and access properties of layers



# Understand *QGIS* interface

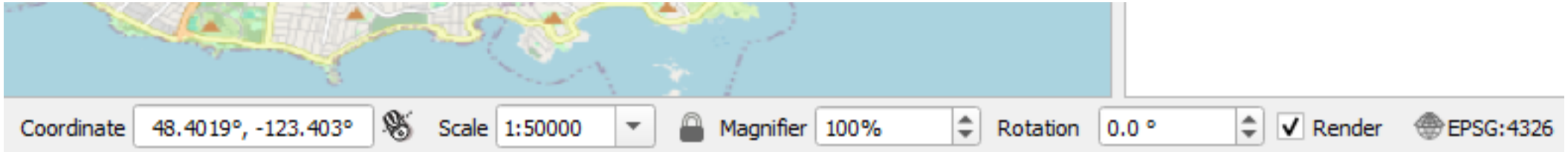



**Map View:**

The central area where spatial data is displayed. Interact with the map, visualize layers and analyze spatial relationships.

**Status Bar:**

Located at the bottom, it provides information about current project: coordinate display, scale and CRS settings.



# Toolbar essentials

New project, open, and save

Pan

zoom

“**zoom full**”

Add data

Proceed to **Workshop Exercise**

There are **Check-In** slides to **ask** for help
(or **ask** sooner!)

# Resources going forward:

**QGIS – additional resources:**

- QGIS Training Manual: https://docs.qgis.org/3.44/en/docs/training_manual/index.html
- QGIS User Guide: https://docs.qgis.org/3.44/en/docs/user_manual/index.html
- QGIS Tutorials & Tips: https://www.qgistutorials.com/

**Find data:**

- GeoSpatial Data Guide: http://libguides.uvic.ca/geospatialdata

**Questions or problems:**

- UVic Geospatial Librarian (danielbm@uvic.ca) or YCW Geospatial Intern (gabriellewade@uvic.ca)

**UVic full semester GIS courses in the Department of Geography:**

- GEOG222 – Intro to Maps and GIS
- GEOG328 – GIS Analysis

**GIS Skills and Mapping Micro-certificate (approx 30 hrs)**

https://continuingstudies.uvic.ca/science-and-the-environment/programs/gis-skills-and-mapping